Separation, Pooling, and Endogenous Timing

A Tullock Contest under Two-sided Asymmetry

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Master’s Thesis Defense

Roadmap

1. Introduction
2. Literature Review
3. Motivation and Benchmark
4. Main Model
5. Extension

Research Question

- How does prize-value correlation affect equilibrium revelation?
- How does it reshape the informed player's timing incentive?
- Can the benchmark and the fully correlated case be unified in one model?

Main message

- Correlation is the single force that moves the game from separation to pooling.
- The fully correlated case is the endpoint $\rho = 1$, not a separate model.

What This Thesis Does

- Replaces the old benchmark with an **iid benchmark**.
- Promotes **partial correlation** to the main model.
- Characterizes separation, pooling, and coexistence.
- Links subgame equilibrium to **endogenous timing**.

Literature and Gap

Strand	What it contributes
Tullock contests	effort-based competition with probabilistic winning
Endogenous timing	strategic value of moving early or late
Signaling	effort can transmit private information
Gap	the transition from iid values to the fully correlated case is not fully characterized

Why an iid Benchmark?

Issue	Old benchmark	iid benchmark
V_U	public constant	random variable
Architecture	disconnected from main model	same structure as main model
Belief channel	weak	still weak at $\rho = 0$
Role	isolated special case	natural left endpoint

- $\rho = 0$: iid benchmark; $\rho = 1$: fully correlated endpoint.
- The benchmark is a reference point, not the main contribution.

Environment and Timing Regimes

Component	Specification
Players	one informed player I, one uninformed player U
Values	$V_I, V_U \in \{v_L, v_H\}$, with $v_H > v_L > 0$
Marginals	$\Pr(V_I = v_H) = \Pr(V_U = v_H) = q$
Contest	Tullock success function

Regime	Interpretation
SS	simultaneous effort choices
UI	uninformed player moves first; informed player responds
IU	informed player moves first; uninformed player responds

Correlation and the Main Mechanism

- A single parameter $\rho \in [0, 1]$ indexes correlation:

$$\Pr(V_I = v_H, V_U = v_H) = q^2 + \rho q(1 - q).$$

Case	Interpretation
$\rho = 0$	iid benchmark
$0 < \rho < 1$	partial correlation
$\rho = 1$	fully correlated endpoint

- With posterior $m(x_I) = \Pr(V_I = v_H \mid x_I)$, the effective prize is

$$\hat{V}_U(m) = v_L + [q + \rho(m - q)](v_H - v_L),$$

- and the best response is

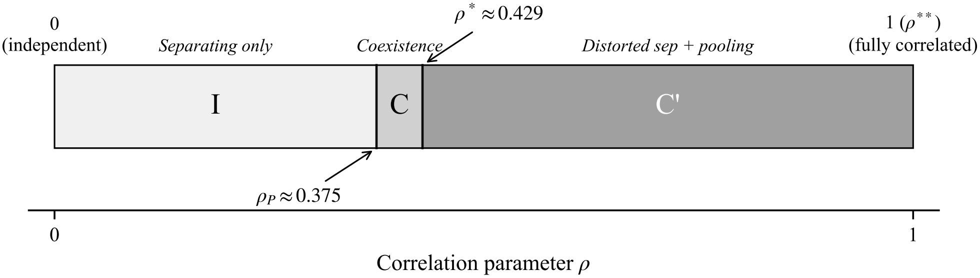
$$x_U^*(x_I, m) = \sqrt{x_I \hat{V}_U(m)} - x_I.$$

Threshold Results

Result	Threshold	Meaning
Undistorted separation	$\rho \leq \rho^*$	survives only below a finite cutoff
Pooling	$\rho \geq \rho_P$	appears before separation fully disappears
Coexistence	$\rho \in [\rho_P, \rho^*]$	separation and pooling coexist

Benchmark calibration $(v_H, v_L, q) = (2, 1, 0.5)$: $\qquad \rho_P = 0.375, \quad \rho^* \approx 0.43.$

Equilibrium Regions in the IU Subgame

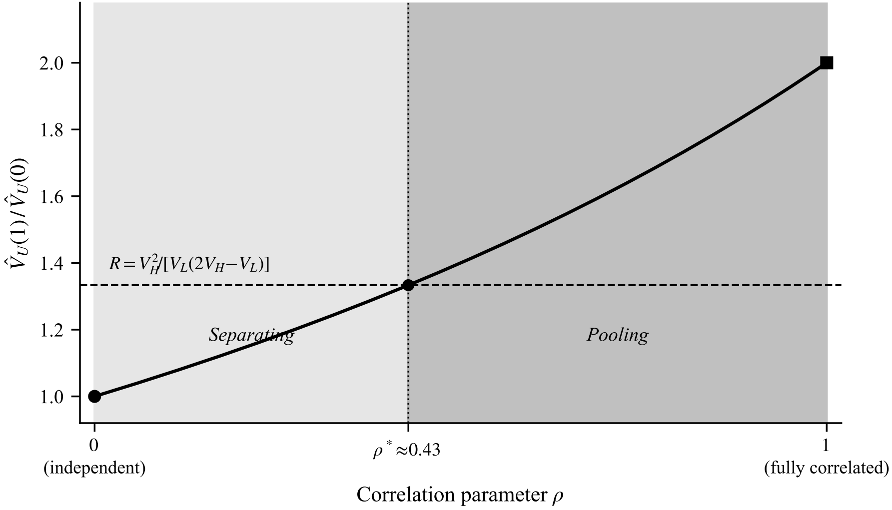


- Region I: unique undistorted separation.
- Region C: separation and pooling coexist.
- Region C’: Riley separation and pooling coexist.
- In the benchmark calibration, the pure-pooling-only region disappears.

Refinement and Timing

Question	Result
Refinement	pooling survives the Intuitive Criterion and D1; D1 selects Riley on the separating side
Timing	$\Delta_I^{IU}(\rho) = \widetilde{\pi}_I^{IU}(\rho) - \widetilde{\pi}_I^{SS}$ decreases with ρ in the separating region
Implication	there is a unique threshold $\rho^\dagger$: low correlation can support informed leadership; high correlation cannot

From Benchmark to Endpoint



- $\rho = 0$: the iid benchmark.

Welfare and Policy Implication

Policy question	Should regulators force simultaneous action?
Comparison object	total effort and rent dissipation across SS, UI, and IU
Why correlation matters	it changes both revelation incentives and timing incentives
Main implication	simultaneity is not automatically welfare-dominant

Extension

Direction	timing chosen after private information is realized
New signal space	a binary timing signal plus a continuous effort signal
Value	richer signaling structure than the current model
Status	left as future work rather than included in incomplete form

Conclusion

- The iid benchmark and the fully correlated case belong to one unified model.
- Correlation is the key parameter behind:
  - separation $\rightarrow$ pooling,
  - multiplicity and refinement,
  - the informed player's timing preference.
- The endpoint result is the limiting case of a broader equilibrium spectrum.